

All are Worth Words: A ViT Backbone for Diffusion Models

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All Are Worth Words: A ViT Backbone for Diffusion Models

Fan Bao, Shen Nie, Kaiwen Xue, Yue Cao, Chongxuan Li, Hang Su, Jun Zhu; Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023, pp. 22669-22679

Abstract

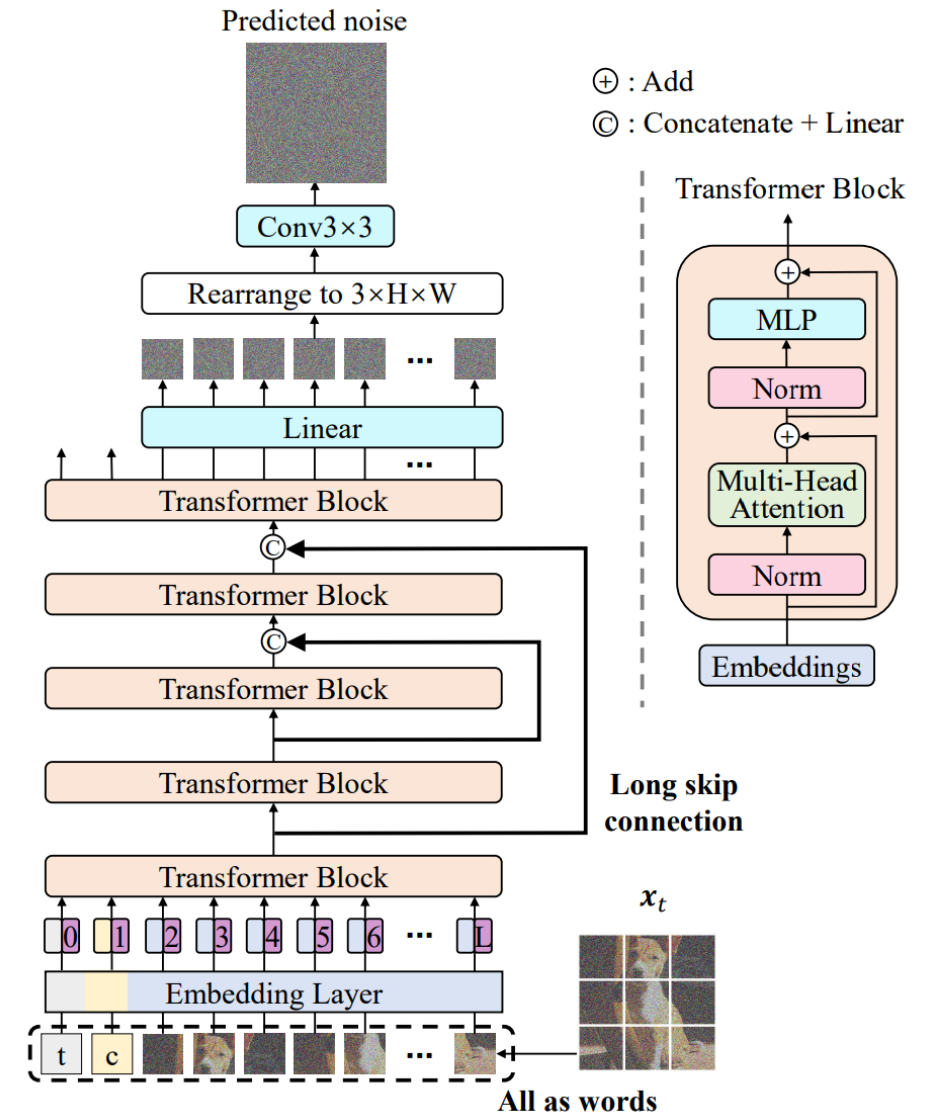
Vision transformers (ViT) have shown promise in various vision tasks while the U-Net based on a convolutional neural network (CNN) remains dominant in diffusion models. We design a simple and general ViT-based architecture (named U-ViT) for image generation with diffusion models. U-ViT is characterized by treating all inputs including the time, condition and noisy image patches as tokens and employing long skip connections between shallow and deep layers. We evaluate U-ViT in unconditional and class-conditional image generation, as well as text-to-image generation tasks, where U-ViT is comparable if not superior to a CNN-based U-Net of a similar size. In particular, latent diffusion models with U-ViT achieve record-breaking FID scores of 2.29 in class-conditional image generation on ImageNet 256x256, and 5.48 in text-to-image generation on MS-COCO, among methods without accessing large external datasets during the training of generative models. Our results suggest that, for diffusion-based image modeling, the long skip connection is crucial while the down-sampling and up-sampling operators in CNN-based U-Net are not always necessary. We believe that U-ViT can provide insights for future research on backbones in diffusion models and benefit generative modeling on large scale cross-modality datasets.



- Introduction
- Method
- Related Work
- Experiments
- Conclusion

- Diffusion models emerge recently for high quality image generation.
 - Applications: text-to-image, image-to-image, video generations, etc
 - And they employ U-Net(CNN) as a backbone.
- On the other hand ViT have shown promise in various vision task.
 - Therefore, a very natural question arises: whether the reliance of the CNN-based U-Net is necessary in diffusion models?
- This paper introduces simple ViT-based architecture called **U-ViT**.

- U-ViT is a simple and general backbone for diffusion models in image generation.
 - They also takes time t , condition c , noised image x_t (as tokens).
 - The long skip connections provide shortcuts for the low-level features.
- U-ViT optionally adds a 3×3 convolutional block before output.
 - This is intended to prevent the potential artifacts in images produced by transformers



- **The way to combine the long skip branch**
 - **(1) Linear(Concat(h_m , h_s))** (2) $h_m + h_s$ (3) $h_m + \text{Linear}(h_s)$ (4) $\text{Linear}(h_m + h_s)$ (5) dropped
- **The way to feed the time into the network**
 - **(1) treat is as a token** (2) incorporate the time after the layer norm in the transformer block.
- **The way to add an extra convolutional block after the transformer**
 - **(1) add a 3×3 conv. after the linear proj.** (2) add a 3×3 conv. before linear proj (3) dropped
- **Variants of the patch embedding**
 - **(1) The original patch embedding adopts a linear proj. that maps a patch to a token embedding.**
 - (2) use a stack of 3×3 conv. followed by a 1×1 conv. to map an image to token embeddings.
- **Variants of the position embedding**
 - **(1) 1D** (2) 2D

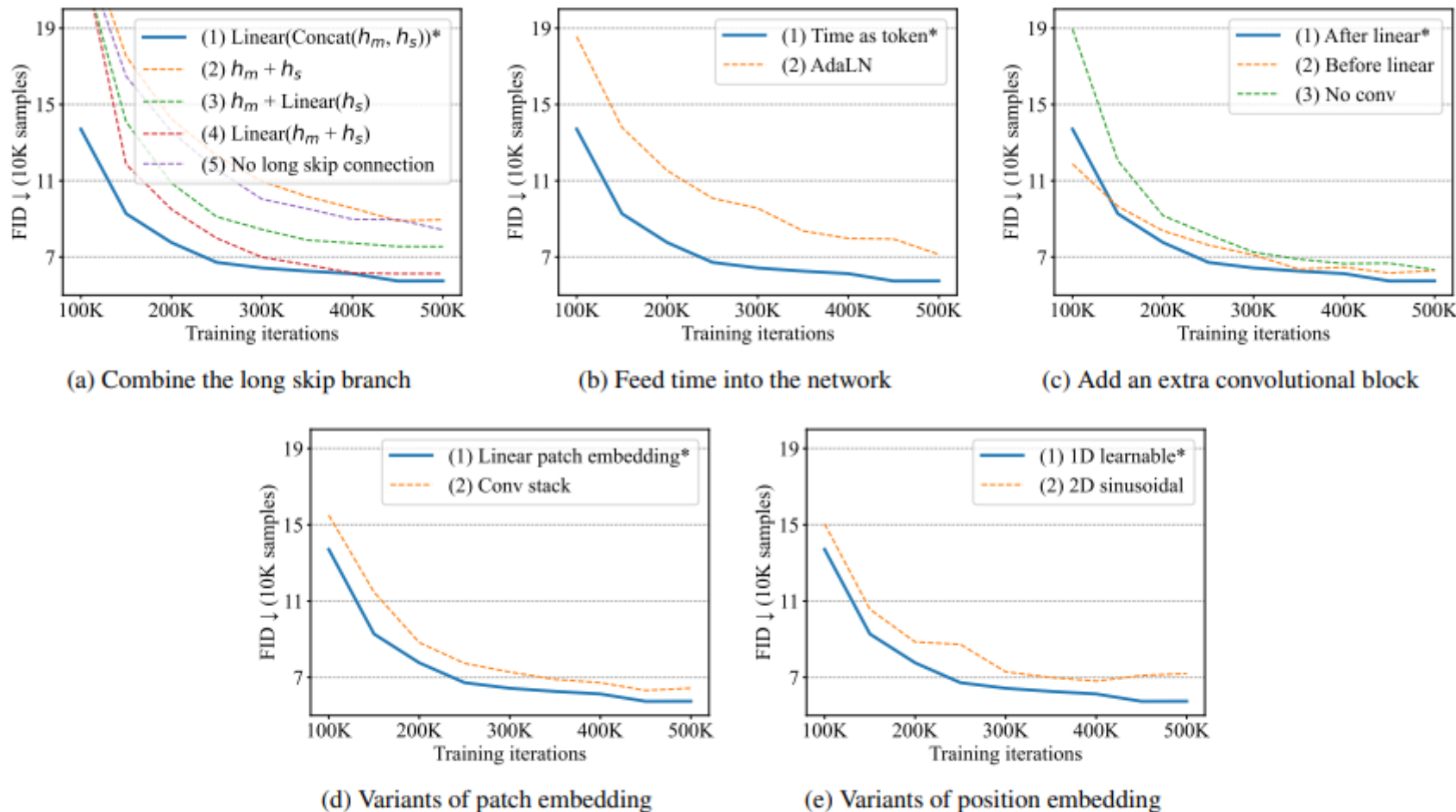
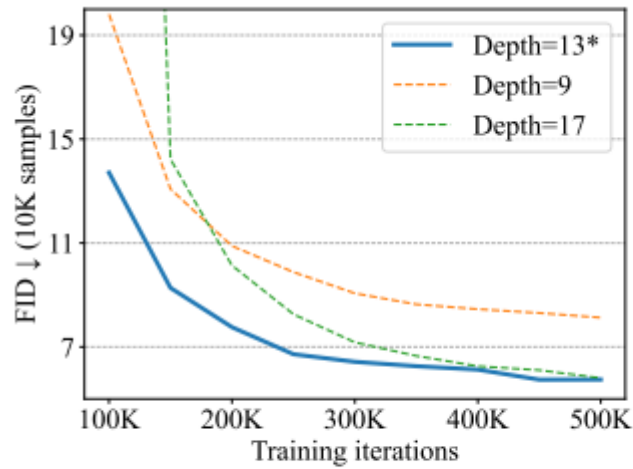
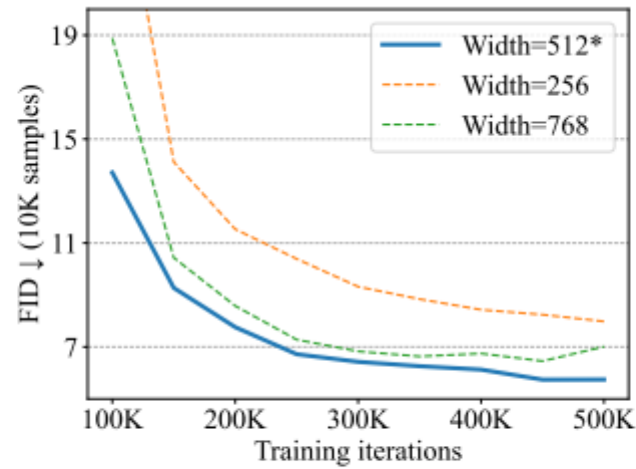


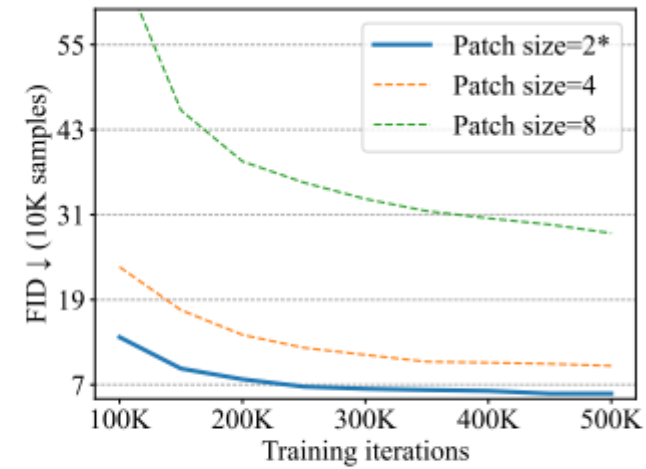
Figure 2. Ablate design choices. The one marked with * is the adopted choice of U-ViT illustrated in Figure 1. Since this ablation aims to determine implementation details, we evaluate FID on 10K generated samples (instead of 50K samples for efficiency).



(a) Depth (#layers)



(b) Width (hidden size)



(c) Patch size

Figure 3. Effect of depth, width and patch size. The one marked with * corresponds to the setting of U-ViT-S/2 (see Table 2).

- Transformers in diffusion models
 - GenViT: long skip connection / incorporates time before norm. layers
 - VQ-Diffusion: obtains a sequence of discrete image tokens via a VQ-GAN
- U-Net in diffusion models
 - initially introduce CNN-based U-Net to model the gradient of log-likelihood function for continuous image data.
 - After that, improvements on the CNN-based U-Net: group norm., multi-head attention, improved residual block and cross attention.
- Improvements of diffusion models
 - Fast sampling, improved training, controllable generation.

- Datasets.
 - For unconditional learning
 - CIFAR10 which contain 50K training images
 - CelebA 64×64, which contain 162,770 training images of human faces.
 - For class-conditional learning
 - ImageNet 64×64, 256×256 and 512×512 resolutions, which contains 1,281,167 training images from 1K different classes.
 - For text-to-image learning
 - MS-COCO at 256×256 resolution, which contains 82,783 training images and 40,504 validation images. Each image is annotated with 5 captions.
- High resolution image generation.
 - LDM for images at 256×256 and 512×512 resolutions. We firstly convert them to latent representations at 32×32 and 64×64 resolutions respectively, using a pretrained image autoencoder provided by Stable Diffusion.
 - Then we model these latent representations using the proposed U-ViT.
- Text-to-image learning.
 - On MS-COCO, we convert discrete texts to a sequence of embeddings using a CLIP text encoder following Stable Diffusion. Then these embeddings are fed into U-ViT as a sequence of tokens.

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Model	#Layers	Hidden size D	MLP size	#Heads	#Params
U-ViT-Small	13	512	2048	8	44M
U-ViT-Small (Deep)	17	512	2048	8	58M
U-ViT-Mid	17	768	3072	12	131M
U-ViT-Large	21	1024	4096	16	287M
U-ViT-Huge	29	1152	4608	16	501M

Table 2. Configurations of U-ViT.

- Training.
 - Optimizer:
 - AdamW with a weight decay of 0.3 for all datasets.
 - learning rate:
 - $2e-4$: most datasets
 - $3e-4$: ImageNet 64×64
 - Iterations:
 - 500K
 - CIFAR10 and CelebA 64×64 with a batch size of 128.
 - ImageNet 512×512 , with a batch size of 1024.
 - 300K
 - ImageNet 64×64 and ImageNet 256×256 .
 - 1M
 - MS-COCO with a batch size of 256.
 - classifier-free guidance:
 - ImageNet 256×256 , ImageNet 512×512 and MS-COCO

Model on CIFAR10		FID ↓
GAN		
StyleGAN2-ADA [31]		2.92
Diff. based on U-Net	#Params	
DDPM [25]	36M	3.17
DDPM [49]	53M	2.90
DDPM++ cont. [67]	62M	2.55
EDM [†] [30]	56M	1.97
Diff. based on ViT	#Params	
GenViT [76]	11M	20.20
U-ViT-S/2	44M	3.11
Model on CelebA 64×64		
FID ↓		
Diff. based on U-Net	#Params	
DDIM [63]	79M	3.26
Soft Truncation [†] [32]	62M	1.90
Diff. model based on ViT	#Params	
U-ViT-S/4	44M	2.87
Model on ImageNet 64×64		
FID ↓		
GAN		
BigGAN-deep [5]		4.06
StyleGAN-XL [59]		1.51
Diff. based on U-Net	#Params	
DDPM (small) [49]	100M	6.92
DDPM (large) [49]	270M	2.92
CDM [26]	Unknown	1.48
ADM [13]	296M	2.07
EDM [†] [30]	296M	1.36
Diff. based on ViT	#Params	
U-ViT-M/4	131M	5.85
U-ViT-L/4	287M	4.26

Model on ImageNet 256×256		FID ↓
GAN		
BigGAN-deep [5]		6.95
StyleGAN-XL [59]		2.30
Discrete diff. based on transformer		
VQ-Diffusion [20]		11.89
VQ-Diffusion (acc0.05) [20]		5.32
Diff. based on U-Net	#Params	
DDPM [49]	270M + 280M (SR)	12.26
CDM [26]	Unknown	4.88
ADM [13]	554M	10.94
ADM-U [13]	296M + 312M (SR)	7.49
ADM-G [13]	554M + 54M (Cls)	4.59
ADM-G, ADM-U [13]	296M + 65M (Cls) + 312M (SR)	3.94
LDM [‡] [54]	400M + 55M (AE)	3.60
Diff. based on ViT	#Params	
U-ViT-H/2 [‡]	501M + 84M (AE)	2.29
Model on ImageNet 512×512		
FID ↓		
GAN		
BigGAN-deep [5]		8.43
StyleGAN-XL [59]		2.41
Diff. based on U-Net	#Params	
ADM [13]	559M	23.24
ADM-U [13]	422M + 309M (SR)	9.96
ADM-G [13]	559M + 54M (Cls)	7.72
ADM-G, ADM-U [13]	422M + 43M (Cls) + 309M (SR)	3.85
Diff. based on ViT	#Params	
U-ViT-H/4 [‡]	501M + 84M (AE)	4.05

	4	5	10	15	20
LDM (trained 178K)	34.48	12.73	4.51	3.87	3.68
U-ViT-H/2 (trained 200K)	16.48	4.94	3.87	3.54	2.91
U-ViT-H/2 (trained 500K)	15.44	4.64	3.18	2.92	2.53

Table 3. FID results on ImageNet 256×256 under different number of sampling steps using the DPM-Solver sampler [44].

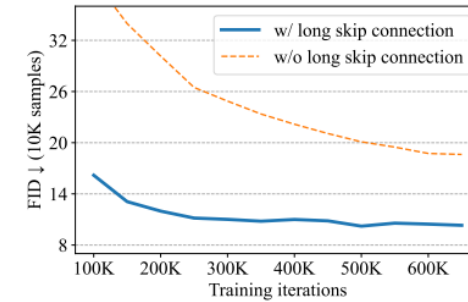


Figure 5. Ablate the long skip connection on ImageNet 256×256 (w/o classifier-free guidance).

Table 1. FID results of unconditional image generation on CIFAR10 and CelebA 64×64, and class-conditional image generation on ImageNet 64×64, 256×256 and 512×512. We mark the best results *among diffusion models* in **bold**. We also include GAN results (gray) for completeness. Methods marked with [†] use advanced training techniques for diffusion models. Methods marked with [‡] model latent representations of images [54] and use classifier-free guidance [27]. We also present the number of parameters of auxiliary components for diffusion models, where SR represents a super-resolution module, AE represents an image autoencoder, and Cls represents a classifier.



Figure 4. Image generation results of U-ViT: selected samples on ImageNet 512×512 and ImageNet 256×256 , and random samples on CIFAR10, CelebA 64×64 , and ImageNet 64×64 .

Model	FID	Type	Training datasets	#Params
Generative model trained on external large dataset (zero-shot)				
DALL-E [53]	~ 28	Autoregressive	DALL-E dataset (250M)	12B
CogView [14]	27.1	Autoregressive	Internal dataset (30M)	4B
LAFITE [82]	26.94	GAN	CC3M (3M)	75M + 151M (TE)
GLIDE [48]	12.24	Diffusion	DALL-E dataset (250M)	3.5B + 1.5B (SR)
Make-A-Scene [19]	11.84	Autoregressive	Union datasets (without MS-COCO) (35M)	4B
DALL-E 2 [52]	10.39	Diffusion	DALL-E dataset (250M)	4.5B + 700M (SR)
Imagen [56]	7.27	Diffusion	Internal dataset (460M) + LAION (400M)	2B + 4.6B (TE) + 600M (SR)
Parti [77]	7.23	Autoregressive	LAION (400M) + FIT (400M) + JFT (4B)	20B + 630M (AE)
Re-Imagen [8]	6.88	Diffusion	KNN-ImageText (50M)	2.5B + 750M (SR)
Generative model trained on external large dataset with access to MS-COCO				
VQ-Diffusion [†] [20]	13.86	Discrete diffusion	Conceptual Caption Subset (7M)	370M
Make-A-Scene [19]	7.55	Autoregressive	Union datasets (with MS-COCO) (35M)	4B
Re-Imagen [‡] [8]	5.25	Diffusion	KNN-ImageText (50M)	2.5B + 750M (SR)
Parti [†] [77]	3.22	Autoregressive	LAION (400M) + FIT (400M) + JFT (4B)	20B + 630M (AE)
Generative model trained on MS-COCO				
AttnGAN [75]	35.49	GAN	MS-COCO (83K)	230M
DM-GAN [83]	32.64	GAN	MS-COCO (83K)	46M
VQ-Diffusion [20]	19.75	Discrete diffusion	MS-COCO (83K)	370M
DF-GAN [70]	19.32	GAN	MS-COCO (83K)	19M
XMC-GAN [79]	9.33	GAN	MS-COCO (83K)	166M
Friro [18]	8.97	Diffusion	MS-COCO (83K)	512M + 186M (TE) + 68M (AE)
LAFITE [82]	8.12	GAN	MS-COCO (83K)	75M + 151M (TE)
U-Net*	7.32	Latent diffusion	MS-COCO (83K)	53M + 123M (TE) + 84M (AE)
U-ViT-S/2	5.95	Latent diffusion	MS-COCO (83K)	45M + 123M (TE) + 84M (AE)
U-ViT-S/2 (Deep)	5.48	Latent diffusion	MS-COCO (83K)	58M + 123M (TE) + 84M (AE)

Table 4. FID results of different models on MS-COCO validation (256×256). U-ViT-S (Deep) increases the number of layers from 13 to 17 compared to U-ViT-S. We also present the number of parameters of auxiliary components for a model when it is reported in the corresponding paper, where SR represents a super-resolution module, AE represents an image autoencoder, and TE represents a text encoder. Methods marked with [†] finetune on MS-COCO. Methods marked with [‡] use MS-COCO as a knowledge base for retrieval. The U-Net* is trained by ourselves to serve as a direct baseline of U-ViT, where we leave other parts unchanged except for the backbone.

Experiments/Text-to-Image Generation on MS-COCO



Figure 6. Text-to-image generation on MS-COCO. All the other settings except the backbone are the same. U-Net and U-ViT generate samples using the same random seed for a fair comparison. The random seed is unselected.

- This work presents U-ViT, a simple and general ViT-based architecture for image generation with diffusion models.
- U-ViT treats all inputs including the time, condition and noisy image patches as tokens and employs long skip connections between shallow and deep layers.
- We evaluate U-ViT in various tasks.
- U-ViT is comparable if not superior to a CNN-based U-Net of a similar size.
- These results suggest that, for diffusion-based image modeling, the long skip connection is crucial while the down/up-sampling operators in CNN-based U-Net are not always necessary.